Supplementary information for the OpenDC-LCA transferability audit

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This supplement documents secondary analyses, software fixtures, and reproducibility details. It must be interpreted with the evidence classes and claim limits in the main manuscript.

S1. Evidence inventory and provenance

The source registry records each provider-native file, version, access URL, declared license or redistribution constraint, local filename, and SHA-256 checksum. Files with uncertain redistribution rights remain under `private-data/` and are excluded from Git. Derived tables are written to both `paper/tables/` and a results directory.

The checksum ledger contains 42 provider files, including all nine historical eGRID downloads. The Applied Energy execution metadata records the files that enter its numerical results, all directly imported analysis modules, the Python runtime, Git commit and dirty-state digest, fixed seeds and iterations, and hashes for every generated table and figure. A provider catalog is not treated as an execution manifest.

The analyses use specific fields from each provider. First, the Microsoft/WSP archive supplies normalized GHG, primary-energy, and blue-water contribution tables, detailed GaBi electricity endpoints, and a pedigree assessment. Second, eGRID supplies state and U.S. generation, CO2, CH4, N2O, and reported CO2e rates. Third, the Federal LCA Commons supplies openLCA process exchanges, product-system links, unit groups, and IPCC AR5-100 characterization factors. Fourth, Boavizta supplies product category, manufacturer, product name, report date, lifetime, total GWP, and manufacturing share. Finally, ÖKOBAUDAT supplies the UUID, product, geography, declared unit, modules A1-A3, GWP, nonrenewable primary energy, and freshwater.

USLCI, GLAD hydrogen, and USGS records are registered and exchange-tested but do not enter a reported cooling LCIA total.

S2. Reproducibility table map

Table S1: Supplementary result map

File	Content	Evidence interpretation
table1_microsoft_reproduction_audit.csv	24 released totals and reconstruction error	Arithmetic consistency only
table2_grid_reductions_vs_air.csv	Relative reductions from the released model	Reconstructed released result
table6_state_rebased_ghg.csv	2023 state screening outputs	Boundary-mismatched numerical transfer
table8_crossover_thresholds.csv	Pairwise crossover values	Conditional numerical landmarks
table9_boavizta_server_records.csv	48 included public product records	Heterogeneous secondary evidence
table14_egrid_historical_state_factors.csv	459 harmonized state-year rates	Direct generation-rate scenarios
table15_egrid_historical_national_factors.csv	Harmonized and provider U.S. series	Historical trend and scope audit
table21_anchor_extrapolation_diagnostic.csv	Within/below/above anchor counts	Transformation-validity diagnostic
table22_boundary_mismatch_stress.csv	Additive allowance cases	Sensitivity, not an upstream estimate

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Table S1: Supplementary result map (continued)

File	Content	Evidence interpretation
table23_joint_assumption_stress.csv	Seeded stress frequencies	Not probability or confidence
table24_functional_unit_sensitivity.csv	Impact-per-service break-even correction	Measurement-resolution target
table25_priority_index_sensitivity.csv	Rank stability across index weights	Heuristic diagnostic
table26_stress_structure_sensitivity.csv	Assumption-block and dependence designs	Design sensitivity, not empirical probability
table28_lifecycle_electricity_factors.csv	71 ordinary at-user consumption systems	Partial provider-linked LCI with documented cutoffs
table29_standardized_rank_robustness.csv	Exact equal-width reversal bounds	Deterministic uncertainty sets
table30_released_endpoint_method_audit.csv	Article/workbook method conflict	Blocks a common-method cooling claim
table31_implied_electricity_response.csv	Slopes implied by released endpoints	Diagnostic response, not measured demand

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Table S1: Supplementary result map (continued)

File	Content	Evidence interpretation
`table32_national_lifecycle_cutoffs.csv`	National unlinked inputs	Omitted-upstream audit
`table33_residual_electricity_factors.csv`	Separate residual-consumption systems	Market-based accounting sensitivity
`table34_boavizta_server_inclusion.csv`	All 55 candidates and reasons	Transparent inclusion/exclusion flow

S3. Released lifecycle results

The 24-cell reconstruction includes GHG, primary energy, and blue water. The main manuscript focuses on GHG because the historical eGRID extension supplies gas-specific air-emission data, not lifecycle primary-energy or electricity-mediated water inventories. The other indicators inherit the released functional unit, boundary, allocation, performance, and electricity cases and were not re-based with eGRID.

S4. Endpoint and crossover diagnostics

The article identifies AR5 GWP100, but the detailed public workbook contains the two numeric electricity endpoints in cells labelled GTP100, while its corresponding GWP100 cells are blank. Direct worksheet-XML parsing verifies that comparative use-phase formulas `Comparative results 0% RE!D118` and `Comparative results 100% RE!D122` point to those blank F29 cells. This unresolved method identity prevents treatment of the cooling screen as a common-method LCA.

Supplementary Figure S1 plots the affine GHG screening lines for air cooling, cold plate, single-phase immersion, and two-phase immersion against 2023 state/DC total-output electricity intensity. The two vertical markers locate the 96.704 kg CO2e/MWh cold-plate/single-phase crossover and the weighted 50-state-plus-DC reference factor.

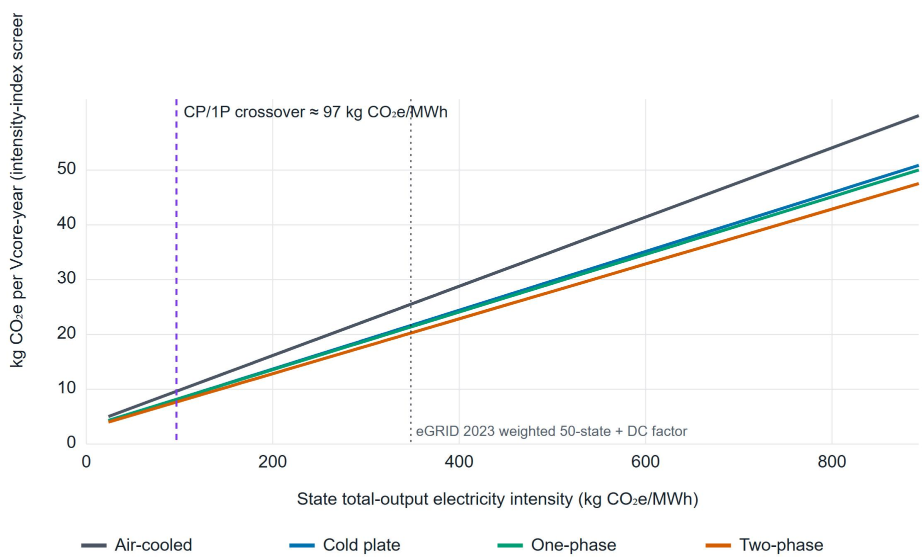

Figure S1: Affine GHG screening lines for air cooling, cold plate, single-phase immersion, and two-phase immersion across 2023 eGRID state/DC total-output rates. The dashed marker locates the 96.704 kg CO2e/MWh cold-plate/single-phase numerical crossover, and the dotted marker locates the weighted 50-state-plus-DC reference factor. The horizontal axis transfers released endpoints across incompatible lifecycle boundaries and is not a background-process substitution.

The two-phase numerical line has no crossover with cold plate or single-phase inside the 2023 state-rate range under the fixed released foreground. This is conditional on server performance, fluid, equipment quantity, lifetime, and the numerical transfer model.

S5. Electricity-system calculation

The 2023 ordinary consumption systems comprise 60 balancing authorities, 10 FERC regions, and the national system. The national partial linked-system result is 422.931 kg CO2e/MWh, with 369.848 from named generation processes and 53.084 from other linked processes. The matched residual-consumption result is 455.350 kg CO2e/MWh and is kept separate.

The national ordinary system solves 582 processes and 1,174 provider links in five fixed-point iterations. It retains 287 unlinked technosphere inputs across 47 processes as explicit cutoffs. Provider scope also omits transmission/distribution infrastructure and some generation infrastructure. The results are partial-scope screening factors, not complete electricity lifecycle benchmarks. The custom solver has an analytic unit-conversion, provider-scaling, elementary-flow-sign, and cutoff test, but independent reproduction in openLCA or Brightway remains a submission action.

S6. Robustness analyses

Supplementary Figure S2 plots two break-even use-phase adjustments against 2023 state electricity intensity. The blue curve gives the cold-plate change required to equal single-phase immersion, while the orange curve gives the two-phase degradation that can occur before another architecture becomes the numerical first rank.

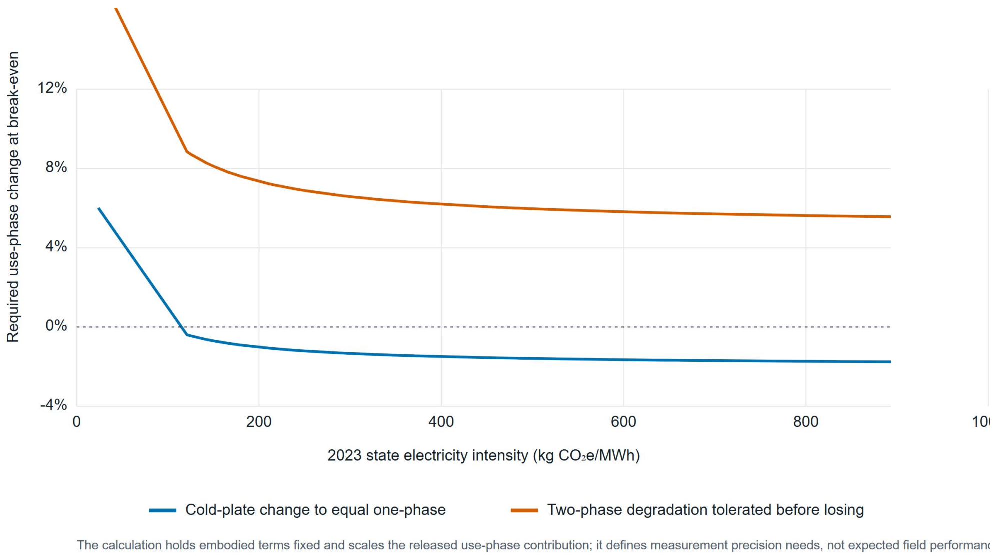

Figure S2: One-at-a-time use-phase changes required to alter close rankings across 2023 state electricity intensities. The blue curve gives the cold-plate adjustment required to equal single-phase immersion, and the orange curve gives the two-phase degradation tolerated before loss of the numerical first rank. These break-even thresholds are experimental discrimination targets, not uncertainty intervals.

At equal relative half-widths, the national first-rank screen reverses at 2.99% for technology-specific use phase, 22.15% for embodied burden, 2.64% for service equivalence, and 1.32% when all four blocks vary. The shared grid-only set does not reverse the order through 99.999%. These are exact set-based certificates around the numerical screen, not physical validation.

The earlier triangular design assigned embodied burden a half-width ten times the use-phase and service widths. Within that declared wide design, embodied-only variation gave a 63.36% two-phase first-rank frequency, compared with 79.52% for use only, 73.90% for service only, and 100% for grid only. The frequencies describe that stress design, not empirical uncertainty. `table27_stress_convergence.csv` records nested 5,000-, 20,000-, and 80,000-draw checks and numerical Monte Carlo standard errors.

S7. Server-product evidence

Supplementary Figure S3 presents the manufacturing-GHG distribution for the 48 included Boavizta server records and marks the 25th percentile, median, and 75th percentile at 1,146, 1,215, and 1,337 kg CO2e/server, respectively.

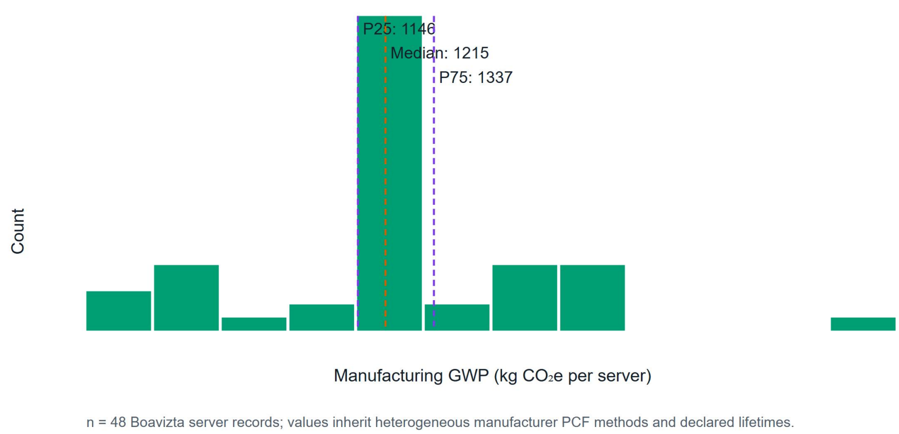

Figure S3: Distribution of manufacturing GHG across 48 included Boavizta server records. The records inherit heterogeneous product rules, hardware configurations, lifetimes, and source methods.

Table 34 provides a candidate-level inclusion audit for all 55

`Datacenter/Server` records by retaining the product identity, total GWP, manufacturing share, lifetime, inclusion status, and exclusion reason. Forty-eight records contain total GWP, manufacturing share, and positive lifetime, while seven Lenovo records are excluded because manufacturing share is blank. The common-scaling stress uses included annualized dispersion but is not a product-substitution model. Decision use requires architecture-specific server count, configuration, useful computation, and lifetime under harmonized product rules.

S8. Construction-material screening

Supplementary Figure S4 compares matched German generic A1-A3 GHG factors for two procurement choices. The selected high-scrap electric-arc-furnace steel route decreases the factor from 2.922 to 1.172 kg CO2e/kg product, while the selected CEM III cement chemistry decreases it from 0.895 to 0.449 kg CO2e/kg product.

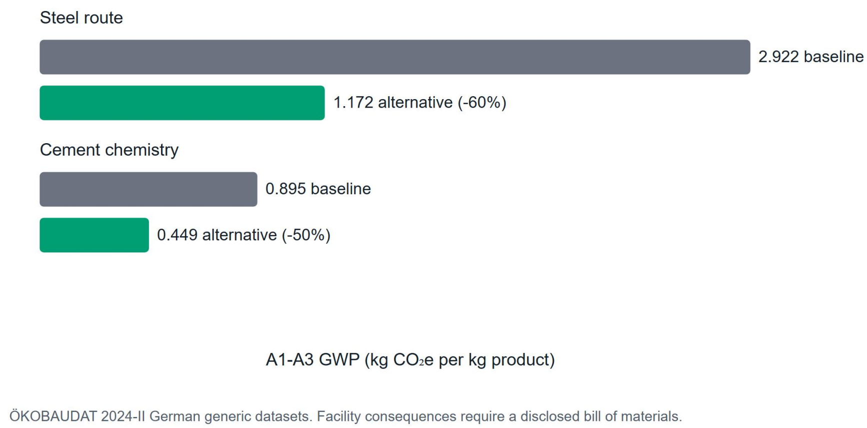

Figure S4: Selected matched ÖKOBAUDAT A1-A3 GHG factors. The comparison identifies per-kilogram procurement leverage and is not propagated to a data-center total.

The selected records are German generic datasets. A facility analysis requires architecture-specific quantities, material grade and supplier, fabrication, transport, replacement, and end-of-life alignment.

S9. Climate and performance-map fixtures

The Fayetteville TMY file contains 8,760 hourly weather records. It is paired with a synthetic performance surface to test schema and unit validation, bounded bilinear interpolation, uncovered-domain rejection, cooling-only partial-PUE integration with one constant grid factor, and interface consistency.

Supplementary Figures S5 and S6 document the two views of this software fixture. Figure S5 integrates the synthetic performance map over 8,760 hourly weather records to compare monthly cooling-only partial PUE for four architectures, while Figure S6 resolves the input surface across dry-bulb temperatures from -15 to 35 °C and IT loads of 40, 70, and 100 kW.

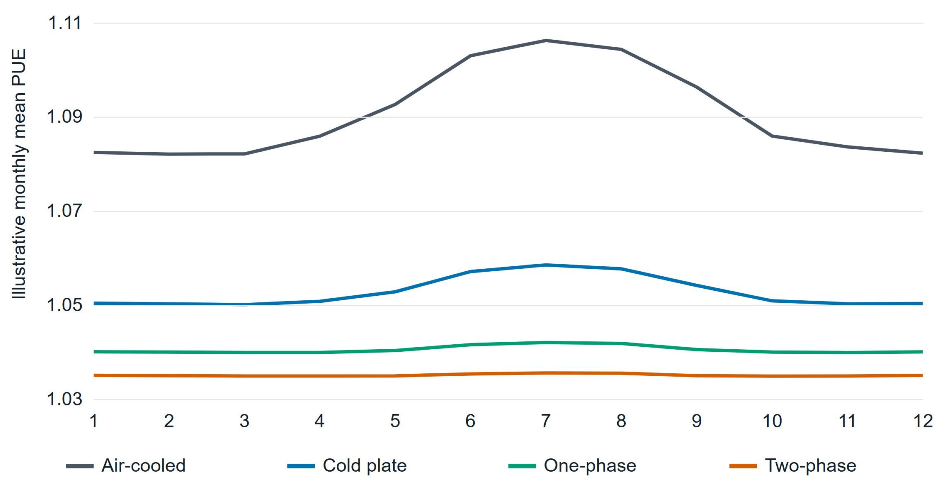

Figure S5: Monthly cooling-only partial PUE for air cooling, cold plate, single-phase immersion, and two-phase immersion obtained by integrating the declared synthetic performance map over 8,760 Fayetteville TMY weather records. The values test the software workflow and do not provide empirical evidence for any cooling technology.

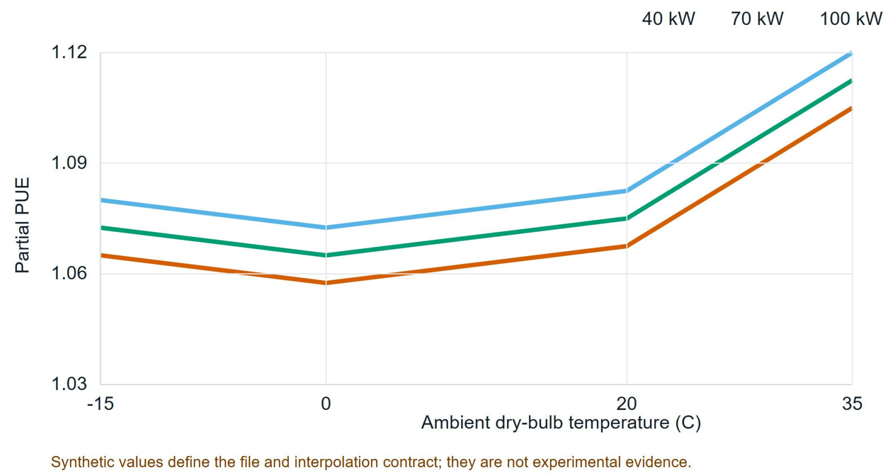

Figure S6: Synthetic partial-PUE surface across dry-bulb temperatures from -15 to 35 °C and IT loads of 40, 70, and 100 kW. The declared grid verifies bounded load-temperature interpolation and does not support a comparative technology assertion.

Wet-bulb temperature is validated in the input schema but is not an interpolation coordinate. Other facility overhead is outside the partial-PUE boundary unless included in the supplied cooling-power field. Submission-grade results require architecture-specific measured power for IT-integral fans, pumps, coolant distribution, heat rejection, controls, and onsite water over a common load-weather domain, plus an energy-balance residual.

S10. Data-priority diagnostic

The base index multiplies mean released GHG contribution by normalized pedigree weakness. Alternative exponents show that use phase is usually, but not always, first. Storage remains in the top tier. This is not formal value of information. Formal analysis would require empirical distributions and correlations, the probability and consequence of a wrong choice, measurement cost and precision, a decision threshold, stakeholder utility, and posterior updating.

S11. Reliability and interoperability

The package supports discrete replacement, linearized replacement, Weibull renewal expectation, and optional Arrhenius acceleration. These functions do not supply architecture-specific reliability without failure, maintenance, duty-cycle, temperature, censoring, and downtime records. Dependent failures, redundancy, and repair queues require an extended model.

Successful openLCA JSON-LD parsing or Brightway mapping establishes syntactic exchange only. Numerical compatibility also requires matching reference product and unit, provider and database version, geography and year, system model, allocation, cutoff, elementary-flow nomenclature, LCIA method, and foreground linkage. USLCI and GLAD archives remain future inputs rather than silent substitutes in the current cooling results.

S12. Software claim controls

Scenario parsing rejects non-finite numeric values. Result manifests retain the field-to-source map and stable digests of source records. Public comparison requests are blocked when functional units, boundaries, electricity accounting, allocation, LCIA methods, or replacement rules differ, or when field-level lineage is missing. A custom comparative boundary requires a structured inclusion/exclusion record that must match across scenarios.

Each custom-boundary energy item is classified as metered, allocated, or excluded. Metered items identify the meter, allocated items identify the allocation method, and excluded items state their rationale. A facility-PUE label requires a complete inventory, a metered IT-load item, and no excluded shared-support item. A dedicated-support ratio remains available for a bounded UPS and dedicated-cooling pathway, but it is not PUE.

These are metadata blockers, not authentication, parameter-level uncertainty mapping, ISO conformity, or evidence that an external review occurred. Useful computation remains a manual prerequisite because the executable functional unit is presently `it_mwh`.

S13. Shared-infrastructure pre-metering case

A preliminary university high-performance-computing site characterization was used to test the custom-boundary record. The site combines i) a UPS-fed IT path, ii) a dedicated liquid-cooling path serving rear-door heat exchangers and GPU-adjacent cooling equipment, and iii) raised-floor room cooling shared with other institutional IT users. The site documentation and tour observations are confidential working records and are not included in this repository.

The site has not supplied historical meter exports, cooling-loop flow records, or workload-service records. It is therefore not an operational validation case, a facility LCA, or a PUE result. The implementation records the required future measurement classes as i) UPS input and output energy, ii) dedicated chiller and pump electricity, iii) chilled-loop supply temperature, return temperature, glycol composition, and flow, iv) shared CRAC energy or an explicit allocation sensitivity, and v) aggregate computing service. This structure keeps directly metered, allocated, and excluded energy distinct until the required evidence exists.

For the eventual liquid-loop assessment, heat removal must be calculated from the measured mass flow, mixture heat capacity, and supply-return temperature difference. Rated refrigeration capacity cannot be substituted for chiller electricity. The first publishable facility result will therefore report a dedicated-support ratio and a shared-cooling allocation range before any facility-PUE claim.

S14. Reproduction commands

Run the following commands from the repository root.

```
python -m pip install -e .
python scripts/run_submission_pipeline.py --skip-refresh
```

Omit `-skip-refresh` to start from provider downloads. Use `-skip-documents` when document-conversion dependencies are unavailable. The pipeline runs source-manifest creation, research, integrated-evidence, and Applied Energy analyses in order, followed by tests and document builds.

A clone without `private-data/` runs the public package tests and explicitly skips the paper-reconstruction tests. Rebuilding the paper requires locally held provider files whose hashes appear in the manifest. Archive the source manifest, execution metadata, derived tables, exact Git commit, compiled main text, and supplement together.